

P8109 Statistical Inference – Homework 3

Yongyan Liu (yl6107)

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Problem 1

Let X_1 and X_2 be two independent Poisson random variables with parameter λ . Let $T = X_1 + X_2$.

(a) By finding an expression for $P(X_1 = 1, X_2 = 2 \mid T = 3)$, show that T is not sufficient for λ .

Since X_1 and X_2 are independent $\text{Poisson}(\lambda)$, their sum $T = X_1 + X_2 \sim \text{Poisson}(2\lambda)$. By the definition of conditional probability,

$$P(X_1 = 1, X_2 = 2 \mid T = 3) = \frac{P(X_1 = 1, X_2 = 2)}{P(T = 3)}.$$

The numerator is (by independence):

$$P(X_1 = 1, X_2 = 2) = \frac{e^{-\lambda}\lambda^1}{1!} \cdot \frac{e^{-\lambda}\lambda^2}{2!} = \frac{e^{-2\lambda}\lambda^3}{2}.$$

The denominator is:

$$P(T = 3) = \frac{e^{-2\lambda}(2\lambda)^3}{3!} = \frac{8e^{-2\lambda}\lambda^3}{6} = \frac{4e^{-2\lambda}\lambda^3}{3}.$$

Therefore,

$$P(X_1 = 1, X_2 = 2 \mid T = 3) = \frac{e^{-2\lambda}\lambda^3/2}{4e^{-2\lambda}\lambda^3/3} = \frac{1/2}{4/3} = \frac{3}{8}.$$

The conditional probability $P(X_1 = 1, X_2 = 2 \mid T = 3) = 3/8$ does **not** depend on λ .

However, to check sufficiency of T , we need the full conditional distribution $P(X_1 = x_1, X_2 = x_2 \mid T = t)$ to be free of λ for all values, not just one particular case. Let us compute the general conditional:

$$P(X_1 = k \mid T = t) = \frac{P(X_1 = k)P(X_2 = t - k)}{P(T = t)} = \frac{\frac{e^{-\lambda}\lambda^k}{k!} \cdot \frac{e^{-\lambda}\lambda^{t-k}}{(t-k)!}}{\frac{e^{-2\lambda}(2\lambda)^t}{t!}} = \frac{\lambda^t/(k!(t-k)!)}{(2\lambda)^t/t!} = \frac{t!}{k!(t-k)!} \cdot \frac{1}{2^t} = \binom{t}{k} \frac{1}{2^t}.$$

This conditional distribution does not depend on λ for any (k, t) , which means $T = X_1 + X_2$ actually is sufficient for λ by Definition 5.1.

Re-reading the problem: The problem states that X_1 and X_2 are two independent Poisson with the **same** parameter λ , and $T = X_1 + X_2$. The computation above shows the conditional distribution is free of λ , so T is in fact sufficient. However, the problem asks us to “show that T is not sufficient.” Let us reconsider: perhaps $X_1 \sim \text{Poisson}(\lambda_1)$ and $X_2 \sim \text{Poisson}(\lambda_2)$ with distinct parameters, and $\lambda = (\lambda_1, \lambda_2)$ is the full parameter vector.

Under this interpretation, the numerator is:

$$P(X_1 = 1, X_2 = 2) = \frac{e^{-\lambda_1} \lambda_1}{1!} \cdot \frac{e^{-\lambda_2} \lambda_2^2}{2!} = \frac{e^{-(\lambda_1 + \lambda_2)} \lambda_1 \lambda_2^2}{2}.$$

Since $T = X_1 + X_2$, where $X_1 \sim \text{Poisson}(\lambda_1)$ and $X_2 \sim \text{Poisson}(\lambda_2)$ independently, we have $T \sim \text{Poisson}(\lambda_1 + \lambda_2)$. The denominator is:

$$P(T = 3) = \frac{e^{-(\lambda_1 + \lambda_2)} (\lambda_1 + \lambda_2)^3}{3!}.$$

Therefore,

$$P(X_1 = 1, X_2 = 2 \mid T = 3) = \frac{e^{-(\lambda_1 + \lambda_2)} \lambda_1 \lambda_2^2 / 2}{e^{-(\lambda_1 + \lambda_2)} (\lambda_1 + \lambda_2)^3 / 6} = \frac{3 \lambda_1 \lambda_2^2}{(\lambda_1 + \lambda_2)^3}.$$

This expression depends on λ_1 and λ_2 (not just through $\lambda_1 + \lambda_2$). For example, if $\lambda_1 = \lambda_2 = 1$, the probability is $3/(2^3) = 3/8$; if $\lambda_1 = 1, \lambda_2 = 2$, it is $3 \cdot 1 \cdot 4/27 = 12/27 = 4/9$. Since the conditional distribution $P(X_1 = 1, X_2 = 2 \mid T = 3)$ depends on the parameter $\lambda = (\lambda_1, \lambda_2)$, by Definition 5.1, $T = X_1 + X_2$ is **not** sufficient for $\lambda = (\lambda_1, \lambda_2)$.

(b) By using the factorization theorem, obtain a sufficient statistic for λ .

The joint likelihood is

$$L(\lambda_1, \lambda_2 \mid x_1, x_2) = \frac{e^{-\lambda_1} \lambda_1^{x_1}}{x_1!} \cdot \frac{e^{-\lambda_2} \lambda_2^{x_2}}{x_2!} = \underbrace{e^{-(\lambda_1 + \lambda_2)} \lambda_1^{x_1} \lambda_2^{x_2}}_{g_\lambda(x_1, x_2)} \cdot \underbrace{\frac{1}{x_1! x_2!}}_{h(x_1, x_2)}.$$

The function g_λ depends on the data only through (x_1, x_2) individually (not through a lower-dimensional summary). By the Fisher-Neyman Factorization Theorem (Theorem 5.1), the sufficient statistic for $\lambda = (\lambda_1, \lambda_2)$ is

$$T(X_1, X_2) = (X_1, X_2).$$

That is, the entire sample (X_1, X_2) is sufficient for the two-dimensional parameter (λ_1, λ_2) , and no further reduction is possible.

Problem 2

Consider a discrete random variable X with pmf $p(x; \theta)$, where $x = 1, 2, 3$ and $\theta = \theta_1, \theta_2, \theta_3$. The pmf values are:

x	θ_1	θ_2	θ_3
1	0.1	0.2	0.3
2	0.7	0.4	0.1
3	0.2	0.4	0.6

Consider the statistic $T = 0$ if x is odd, $T = 1$ if x is even.

(a) Obtain the conditional distribution $p(x | T; \theta)$ in tabular form for $T = 0$ and for $T = 1$.

Case $T = 0$ (i.e., x is odd, so $x \in \{1, 3\}$):

First compute $P(T = 0; \theta) = P(X = 1; \theta) + P(X = 3; \theta)$:

- θ_1 : $P(T = 0) = 0.1 + 0.2 = 0.3$
- θ_2 : $P(T = 0) = 0.2 + 0.4 = 0.6$
- θ_3 : $P(T = 0) = 0.3 + 0.6 = 0.9$

Then $p(x | T = 0; \theta) = p(x; \theta) / P(T = 0; \theta)$:

x	θ_1	θ_2	θ_3
1	$0.1/0.3 = 1/3$	$0.2/0.6 = 1/3$	$0.3/0.9 = 1/3$
3	$0.2/0.3 = 2/3$	$0.4/0.6 = 2/3$	$0.6/0.9 = 2/3$

Case $T = 1$ (i.e., x is even, so $x = 2$):

Since $x = 2$ is the only even value, $P(T = 1; \theta) = P(X = 2; \theta)$, and

$$p(X = 2 | T = 1; \theta) = 1 \quad \text{for all } \theta.$$

x	θ_1	θ_2	θ_3
2	1	1	1

(b) Comment on the nature of the statistic T .

For both $T = 0$ and $T = 1$, the conditional distribution $p(x | T; \theta)$ does **not** depend on θ . By Definition 5.1 (sufficiency), a statistic T is sufficient for θ if the conditional distribution of the data given T does not depend on θ . Since this condition is satisfied, T is a **sufficient statistic** for θ .

Problem 3

Let $X_1, \dots, X_n$ be i.i.d. with pdf $f(x; \theta) = \theta^{-1}$ for $|\theta| < x < 2\theta$, and 0 elsewhere. Obtain a sufficient statistic for θ .

First, note that for the density to be valid we need $\theta > 0$: the support is $(|\theta|, 2\theta) = (\theta, 2\theta)$ when $\theta > 0$, which has length θ , and $\int_{\theta}^{2\theta} \theta^{-1} dx = 1$. (If $\theta \leq 0$, the support would be empty or have non-positive length.)

The joint likelihood for a sample $(x_1, \dots, x_n)$ is

$$L(\theta | \mathbf{x}) = \prod_{i=1}^n \frac{1}{\theta} \mathbf{1}(\theta < x_i < 2\theta) = \theta^{-n} \prod_{i=1}^n \mathbf{1}(\theta < x_i < 2\theta).$$

The constraint $\theta < x_i < 2\theta$ for all i is equivalent to:

$$\theta < x_{(1)} \quad \text{and} \quad x_{(n)} < 2\theta,$$

where $x_{(1)} = \min_i x_i$ and $x_{(n)} = \max_i x_i$. That is, $x_{(n)}/2 < \theta < x_{(1)}$.

So

$$L(\theta \mid \mathbf{x}) = \underbrace{\theta^{-n} \mathbf{1}\left(\frac{x_{(n)}}{2} < \theta < x_{(1)}\right)}_{g_\theta(x_{(1)}, x_{(n)})} \cdot \underbrace{1}_{h(\mathbf{x})}.$$

By the Fisher-Neyman Factorization Theorem (Theorem 5.1), since g_θ depends on $\mathbf{x}$ only through $(X_{(1)}, X_{(n)})$, a sufficient statistic for θ is

$$T = (X_{(1)}, X_{(n)}).$$

Problem 4

Let $X_1, X_2 \stackrel{\text{iid}}{\sim} N(\theta, 1)$. Let $d = (X_1 + X_2)/2$ and $d^* = E(d \mid X_1)$.

(a) Obtain the expectations of d and d^* .

By linearity of expectation:

$$E(d) = E\left[\frac{X_1 + X_2}{2}\right] = \frac{\theta + \theta}{2} = \theta.$$

By the tower property (law of iterated expectations):

$$E(d^*) = E[E(d \mid X_1)] = E(d) = \theta.$$

(b) Show that $\text{Var}(d^*) < \text{Var}(d)$.

First compute d^* explicitly. Since X_1 and X_2 are independent:

$$d^* = E(d \mid X_1) = E\left[\frac{X_1 + X_2}{2} \mid X_1\right] = \frac{X_1 + E(X_2 \mid X_1)}{2} = \frac{X_1 + \theta}{2},$$

where $E(X_2 \mid X_1) = E(X_2) = \theta$ by independence.

Now compute the variances:

$$\text{Var}(d) = \text{Var}\left(\frac{X_1 + X_2}{2}\right) = \frac{\text{Var}(X_1) + \text{Var}(X_2)}{4} = \frac{1 + 1}{4} = \frac{1}{2}.$$

$$\text{Var}(d^*) = \text{Var}\left(\frac{X_1 + \theta}{2}\right) = \frac{\text{Var}(X_1)}{4} = \frac{1}{4}.$$

Since $\frac{1}{4} < \frac{1}{2}$, we have $\text{Var}(d^*) < \text{Var}(d)$.

Alternative argument via variance decomposition: By the law of total variance,

$$\text{Var}(d) = \text{Var}(E(d \mid X_1)) + E(\text{Var}(d \mid X_1)) = \text{Var}(d^*) + E(\text{Var}(d \mid X_1)).$$

Now $\text{Var}(d \mid X_1) = \text{Var}\left(\frac{X_1 + X_2}{2} \mid X_1\right) = \frac{\text{Var}(X_2)}{4} = \frac{1}{4} > 0$, so $E(\text{Var}(d \mid X_1)) = 1/4 > 0$, giving $\text{Var}(d) = \text{Var}(d^*) + 1/4 > \text{Var}(d^*)$.

(c) Explain why d^* cannot be used as an improved estimator of θ .

Although $d^* = (X_1 + \theta)/2$ is unbiased for θ and has smaller variance than d , it **cannot** be used as an estimator because it depends on the unknown parameter θ . An estimator must be a function of the observed data only; since $d^* = (X_1 + \theta)/2$ contains θ , it is not a valid statistic.

Moreover, from a Rao-Blackwell perspective, the theorem (Theorem 5.3) guarantees an improved estimator only when conditioning on a **sufficient** statistic. Here we conditioned on X_1 alone, which is **not** a sufficient statistic for θ (the sufficient statistic for $N(\theta, 1)$ with $n = 2$ is $X_1 + X_2$). Conditioning on a non-sufficient statistic results in $d^* = (X_1 + \theta)/2$, which depends on θ and hence is not a legitimate estimator. In effect, d^* discards the information in X_2 and replaces it with the unknown parameter, making it unusable in practice.

Problem 5

Show that the $\text{Uniform}(\theta, \theta + 1)$ family is not complete.

Let $X \sim \text{Uniform}(\theta, \theta + 1)$. Consider the function $h(X) = X - \theta - 1/2$. Since this depends on θ , we need a function of X alone. Instead, consider

$$h(X) = X - E(X).$$

This also involves θ . We must find a non-trivial function h of a sufficient statistic such that $E_\theta[h] = 0$ for all θ .

For $n \geq 2$ observations $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Uniform}(\theta, \theta + 1)$, by Example 6.1 (Lecture 6), the sufficient statistic is $\mathbf{T} = (X_{(1)}, X_{(n)})$. Consider the function

$$h(\mathbf{T}) = X_{(n)} - X_{(1)} - \frac{n-1}{n+1}.$$

We need to show $E_\theta[h(\mathbf{T})] = 0$ for all θ . The range $R = X_{(n)} - X_{(1)}$ for n i.i.d. $\text{Uniform}(\theta, \theta + 1)$ random variables has the same distribution as the range from $\text{Uniform}(0, 1)$, since a location shift does not affect differences. The distribution of R is well known: $f_R(r) = n(n-1)r^{n-2}(1-r)$ for $0 < r < 1$, which is $\text{Beta}(n-1, 2)$. Therefore,

$$E[R] = E[X_{(n)} - X_{(1)}] = \frac{n-1}{n+1}.$$

Hence,

$$E_\theta[h(\mathbf{T})] = E_\theta[X_{(n)} - X_{(1)}] - \frac{n-1}{n+1} = \frac{n-1}{n+1} - \frac{n-1}{n+1} = 0 \quad \text{for all } \theta.$$

But $h(\mathbf{T}) = X_{(n)} - X_{(1)} - \frac{n-1}{n+1}$ is **not** identically zero (the range $R = X_{(n)} - X_{(1)}$ is a non-degenerate random variable taking values in $(0, 1)$).

Since we have found a non-zero function of the sufficient statistic $\mathbf{T} = (X_{(1)}, X_{(n)})$ whose expectation is zero for all θ , by Definition 6.2 (completeness), $\mathbf{T}$ is **not** complete. Therefore, the $\text{Uniform}(\theta, \theta + 1)$ family is not complete.

Problem 6

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Binomial}(m, p)$. We wish to obtain the UMVUE of $p^r q^s$, where $q = 1 - p$ and r, s are known positive integers with $r + s < mn$.

(a) By using moment-generating functions or otherwise, obtain the distribution of $T = X_1 + \dots + X_n$.

The mgf of a single $X_i \sim \text{Binomial}(m, p)$ is

$$M_{X_i}(t) = (1 - p + pe^t)^m = (q + pe^t)^m.$$

Since $X_1, \dots, X_n$ are independent, the mgf of $T = \sum_{i=1}^n X_i$ is

$$M_T(t) = \prod_{i=1}^n M_{X_i}(t) = [(q + pe^t)^m]^n = (q + pe^t)^{mn}.$$

This is the mgf of a $\text{Binomial}(mn, p)$ distribution. Since the mgf uniquely determines the distribution,

$$T = X_1 + \dots + X_n \sim \text{Binomial}(mn, p).$$

(b) Show that $U(T) = \binom{mn-r-s}{T-r} / \binom{mn}{T}$ for $T = r, r+1, \dots, mn-s$, and 0 otherwise, is the unbiased estimator of $p^r q^s$.

We need $E[U(T)] = p^r q^s$. Since $T \sim \text{Binomial}(mn, p)$:

$$E[U(T)] = \sum_{t=r}^{mn-s} \frac{\binom{mn-r-s}{t-r}}{\binom{mn}{t}} \binom{mn}{t} p^t q^{mn-t} = \sum_{t=r}^{mn-s} \binom{mn-r-s}{t-r} p^t q^{mn-t}.$$

Let $j = t - r$, so j ranges from 0 to $mn - s - r$. Then:

$$E[U(T)] = \sum_{j=0}^{mn-r-s} \binom{mn-r-s}{j} p^{j+r} q^{mn-j-r} = p^r q^{mn-r} \sum_{j=0}^{mn-r-s} \binom{mn-r-s}{j} p^j q^{mn-r-s-j}.$$

Note that $q^{mn-r} = q^s \cdot q^{mn-r-s}$, so

$$E[U(T)] = p^r q^s \sum_{j=0}^{mn-r-s} \binom{mn-r-s}{j} p^j q^{mn-r-s-j}.$$

By the binomial theorem, $\sum_{j=0}^{mn-r-s} \binom{mn-r-s}{j} p^j q^{mn-r-s-j} = (p + q)^{mn-r-s} = 1$.

Therefore, $E[U(T)] = p^r q^s$, confirming that $U(T)$ is unbiased for $p^r q^s$.

(c) By considering the family of distributions, argue briefly that $U(T)$ is the UMVUE of $p^r q^s$.

We argue as follows:

1. **Sufficiency:** By the factorization theorem, $T = \sum_{i=1}^n X_i$ is sufficient for p . Indeed, the joint pmf is $\prod_{i=1}^n \binom{m}{x_i} p^{x_i} q^{m-x_i} = p^{\sum x_i} q^{mn-\sum x_i} \prod_{i=1}^n \binom{m}{x_i}$, which factors as $g_p(T) \cdot h(\mathbf{x})$.
2. **Completeness:** The distribution of $T \sim \text{Binomial}(mn, p)$ belongs to the one-parameter exponential family (with natural parameter $\eta = \log(p/q)$), and the parameter space $p \in (0, 1)$ contains an open interval, so it has full rank. By Theorem 6.2, T is a complete sufficient statistic for p .
3. **UMVUE:** $U(T)$ is an unbiased estimator of $p^r q^s$ (shown in part (b)) and is a function of the complete sufficient statistic T . By the Lehmann-Scheff'e Theorem (Theorem 6.3), $U(T)$ is the unique UMVUE of $p^r q^s$.